Forward passes by hand

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Image-pattern Network Investigation

THE CONNECTED WEEK

One readable forward-debugging record.

Separate expected/actual values,repair the first invalid step and retain discriminating tests.

Keep one habit: W1–W3/W5 → selected new audit → project regression record → changed-case return.

MAIN LESSON · SECTION 1

Recall · Retrieve before notes

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Composing layers. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Bias terms and decision boundaries. Give an example and a mistake that the idea helps you avoid.

- Use this reference for the original W1–W3 and W5 exercises. The separately named independent case supplies its own changed facts; use those facts for that attempt before feedback.

- Use a new one-**input** network with two hidden units. Its scores are z1 = 2x - 1 and z2 = -x + 2. Apply ReLU to obtain h1 and h2. The **output** is y = h1 - 2h2 + 1. Parameters stay fixed while the **input** changes.

  The **input** x is an abstract number; these are arithmetic examples, not measurements of children.

- A forward pass evaluates this network in **dependency** order. Compute both scores from x, then both activations, then the **output** contributions h1 and -2h2, and finally add bias 1.

  A **trace** table makes an omitted bias or a sign error easier to locate than a final answer alone.

- At x = 1 the **output** is exactly zero. If we later declare class 1 for y >= 0, equality belongs to class 1. Applying ReLU in the hidden layer does not make the final **output** nonnegative.

Composing layers

My remembered idea, example and correction

Bias terms and decision boundaries

My remembered idea, example and correction

Lesson reference for workbook practice

x	z1 and z2	h1 and h2	y
0	-1 and 2	0 and 2	-3
1	1 and 1	1 and 1	0
3	5 and -1	5 and 0	6

MAIN LESSON · SECTION 5

W1 · Explain the current example

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Main network z[2x−1, −x+2], hReLU each, y=h1−2h2+1. x0:z[−1, 2], h[0, 2], y−3; x.5:z[0, 1.5], h[0, 1.5], y−2; x1:z[1, 1], h[1, 1], y0; x2:z[3, 0], h[3, 0], y4; x3:z[5, −1], h[5, 0], y6. W2 submittedx3y5 omits output bias; z/hare correct. x1 equalhiddenvalues cannot reveal a swap; mainx 0 swappedh[2, 0]wouldgivey 3. Separate scalar picturez 2x−1, hReLU, y−h+4:x0y4, x2y1; submittedx 0h−1 firstfails atactivation.

  Numerical inputs assumed; no task labels or accuracy supplied. Independent model has different formulas.

- **Trace** x = -1 through the lesson network. Record z, h and y.

My reasoning and evidence

MAIN LESSON · SECTION 6

W2 · Trace the calculation

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Main network z[2x−1, −x+2], hReLU each, y=h1−2h2+1. x0:z[−1, 2], h[0, 2], y−3; x.5:z[0, 1.5], h[0, 1.5], y−2; x1:z[1, 1], h[1, 1], y0; x2:z[3, 0], h[3, 0], y4; x3:z[5, −1], h[5, 0], y6. W2 submittedx3y5 omits output bias; z/hare correct. x1 equalhiddenvalues cannot reveal a swap; mainx 0 swappedh[2, 0]wouldgivey 3. Separate scalar picturez 2x−1, hReLU, y−h+4:x0y4, x2y1; submittedx 0h−1 firstfails atactivation.

  Numerical inputs assumed; no task labels or accuracy supplied. Independent model has different formulas.

- At x = 3 a **trace** gives z = [5, -1], h = [5, 0] and y = 5. Identify the first incorrect stage and repair it.

My reasoning and evidence

MAIN LESSON · SECTION 7

W3 · Repair a misconception

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Main network z[2x−1, −x+2], hReLU each, y=h1−2h2+1. x0:z[−1, 2], h[0, 2], y−3; x.5:z[0, 1.5], h[0, 1.5], y−2; x1:z[1, 1], h[1, 1], y0; x2:z[3, 0], h[3, 0], y4; x3:z[5, −1], h[5, 0], y6. W2 submittedx3y5 omits output bias; z/hare correct. x1 equalhiddenvalues cannot reveal a swap; mainx 0 swappedh[2, 0]wouldgivey 3. Separate scalar picturez 2x−1, hReLU, y−h+4:x0y4, x2y1; submittedx 0h−1 firstfails atactivation.

  Numerical inputs assumed; no task labels or accuracy supplied. Independent model has different formulas.

- **Trace** x = 0.5 and x = 2. Which hidden score is zero in each case?

> **My reasoning and evidence**

MAIN LESSON · SECTION 9

W5 · Explain the limitation

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Main network z[2x−1, −x+2], hReLU each, y=h1−2h2+1. x0:z[−1, 2], h[0, 2], y−3; x.5:z[0, 1.5], h[0, 1.5], y−2; x1:z[1, 1], h[1, 1], y0; x2:z[3, 0], h[3, 0], y4; x3:z[5, −1], h[5, 0], y6. W2 submittedx3y5 omits output bias; z/hare correct. x1 equalhiddenvalues cannot reveal a swap; mainx 0 swappedh[2, 0]wouldgivey 3. Separate scalar picturez 2x−1, hReLU, y−h+4:x0y4, x2y1; submittedx 0h−1 firstfails atactivation.

  Numerical inputs assumed; no task labels or accuracy supplied. Independent model has different formulas.

- Explain why testing only x = 1 in the lesson model may miss a hidden-order swap.

> **My reasoning and evidence**

MAIN LESSON · SECTION 4

Try alone · Selected transfer

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use z1 = x + 1, z2 = 2 - 2x, h = ReLU(z) component by component, and y = 2h1 + h2 - 3. **Trace** x = 0 and x = 2. A submitted x = 2 **trace** uses h = [3, -2] and y = 1. Find its first error and correct all later values.

- Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Independent case: Audit a different network independently

> **My independent explanation, trace and limits**

MAIN LESSON · SECTION 10

Project · Keep the mathematical evidence

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Add a labelled forward passes by hand **evidence** sheet to Image-pattern Network Investigation. Show how you can record scores and activations in **dependency** order.

  Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a **trace** is insufficient.

> **My project change, supporting evidence and remaining limit**

MAIN LESSON · SECTION 11

Feedback · Repair and return

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use W2: Locate a bias omission for Forward passes by hand. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

> **The step I completed without help, and the hint I needed**

▼ Extra practice · Individual reasoning check

REFERENCE · SECTION 3

Extra practice · Individual reasoning check

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use the worked case for Forward passes by hand. Proposed explanation: "ReLU means the **output** cannot be negative". Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Earlier W4 response copy

REFERENCE · SECTION 8

Reference · Earlier W4 response copy

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use z1 = x + 1, z2 = 2 - 2x, h = ReLU(z) component by component, and y = 2h1 + h2 - 3. **Trace** x = 0 and x = 2. A submitted x = 2 **trace** uses h = [3, -2] and y = 1. Find its first error and correct all later values.

My reasoning and evidence

▼ **Extra practice · Pictured case**

REFERENCE · SECTION 12

Extra practice · Pictured case

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Audit a forward pass by writing every value at its correct stage.
- Toy scalar network z=2x-1, h=ReLU(z), y=-h+4. **Trace** x=0 and x=2.
- All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.
- Why should feedback identify h before correcting only y? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

- Support used: alone / hint / worked example. This record is practice **evidence**.
- Project connection: For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

  Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

AI ACADEMY / AI-5/6

Forward passes by hand

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Preactivation

At x=0, z=-1; at x=2, z=3.



2. Activation

At x=0, h=0; at x=2, h=3.



3. Output

Outputs are 4 and 1 respectively.

CHANGE ONE THING

A submitted x = 0 trace uses h=-1 and y = 5.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

My picture-based explanation and changed-case prediction

▼ **Reference · Project milestone**

REFERENCE · SECTION 13

Reference · Project milestone

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Project: Image-pattern Network Investigation
- Current milestone: Frame the need. Name the user, the problem, allowed **input**s and a safe simple starting method.
- For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

  Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	

Evidence to keep	My record
One change since the previous checkpoint and why	
One error or limit and the next test	

> **Explain what this week adds to your project. Reuse last week's record rather than starting again.**

▼ **Extra practice · Predict,check and repair**

REFERENCE · SECTION 14

Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

First supported question and optional hint

- Audit a forward pass by writing every value at its correct stage.
- Which labelled step matches this detail? **Output**s are 4 and 1 respectively.
- 1. **Preactivation**
- 2. **Output**
- 3. Activation

- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check after the first attempt**

After your attempt, compare with the supported explanation: **Output Output**s are 4 and 1 respectively. A forward pass applies a fixed network to an **input**; it does not update weights.

Keep the **preactivation** z, activated value h and final **output** y in separate **trace** columns.
For zero, ReLU changes minus one to zero.

Using the earlier minus one in the **output** layer is the first error, even if later arithmetic follows that mistaken value correctly.

Changed-case question and optional hint

- Changed-case practice: A submitted x = 0 **trace** uses h=-1 and y = 5. Which conclusion is supported by the supplied facts?

- 1. Its first error is retaining the negative value after ReLU; correct h = 0 gives y = 4.

- 2. This one example guarantees correct results for every future **input**.

- 3. We can decide the result without examining the changed **input** or the procedure.

- Hint if needed: Use the changed condition and the deciding step in the pictorial **trace**. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the changed attempt**

After your attempt, compare with the supported explanation: Its first error is retaining the negative value after ReLU; correct h = 0 gives y = 4. ReLU(-1)=0, so the **output** must use 0, not the earlier **preactivation**.

Repair and retain the record

Review Composing layers (ai-5/5). Goal: **Trace** how one layer supplies the next layer **input**s. Short practice: What intermediate value explains the changed **output**?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Bias terms and decision boundaries (ai-5/3). Goal: Calculate a weighted score with a bias term. Short practice: Is B positive under the original or changed model?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a **trace** rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

▼ Earlier wording, alternatives and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Predict, run and debug

Keep your completed work

▼ Reference · Six-picture response sheets

REFERENCE · SECTION 15

Reference · Six-picture response sheets

> Choose the picture that helps explain the current step.

01. Forward passes by hand: the case

AI ACADEMY / AI-5/6 · 01 · LOOK AT THE FACTS

Forward passes by hand: the case

The situation

Audit a forward pass by writing every value at its correct stage.

Supplied evidence

Toy scalar network z=2x-1, h=ReLU(z), y=-h+4. Trace x=0 and x=2.
All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

By the end, I can explain forward passes by hand using a traced example and a stated limitation.

Follow the labelled picture with the matching explanation. Open the vector diagram

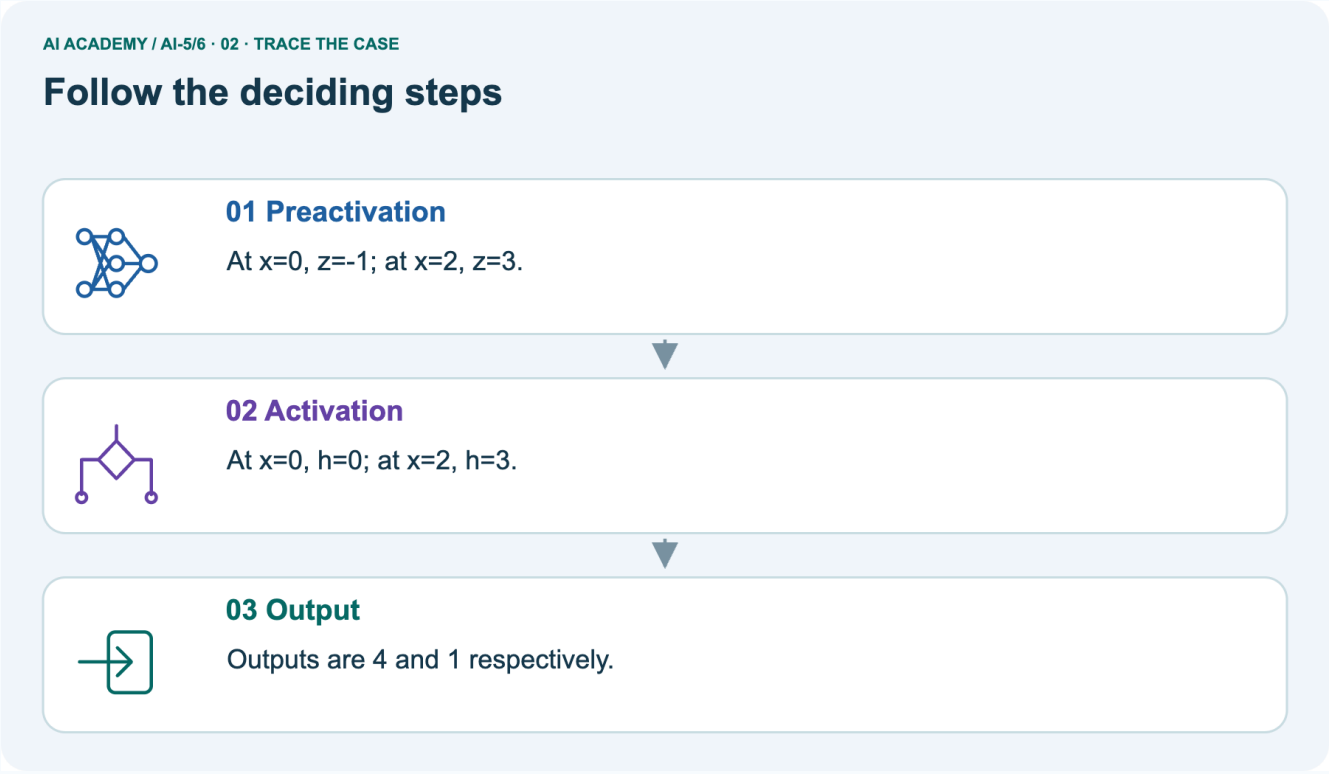
Audit a forward pass by writing every value at its correct stage. Supplied facts: Toy scalar network z=2x-1, h=ReLU(z), y=-h+4. **Trace** x=0 and x=2.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Preactivation: At x=0, z=-1; at x=2, z=3. → Activation: At x=0, h=0; at x=2, h=3. → **Output**: **Output**s are 4 and 1 respectively.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Correct a trace from the earliest error

AI ACADEMY / AI-5/6 · 03 · INSPECT THE LESSON MODEL

Correct a trace from the earliest error

Trace stage at x is 0	Expected value	Submitted value
Scores	[-1, 2]	[-1, 2]
ReLU activations	[0, 2]	[-1, 2]
Output	-3	-4

The columns compare expected and submitted traces. The first disagreement is the activation of -1, not the final addition. Recompute all dependent values after repairing it.

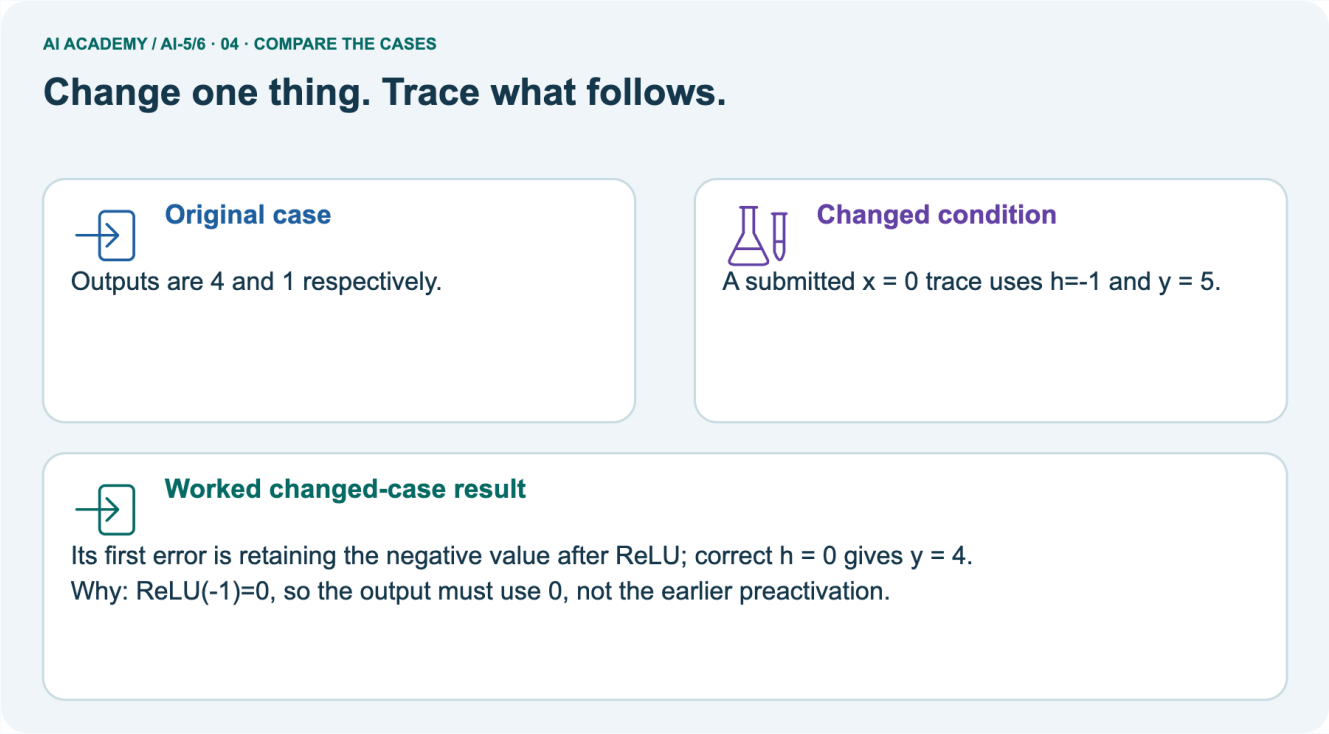
Follow the labelled picture with the matching explanation. Open the vector diagram

Correct a **trace** from the earliest error. Scores | [-1, 2] | [-1, 2]. ReLU activations | [0, 2] | [-1, 2]. **Output** | -3 | -4

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

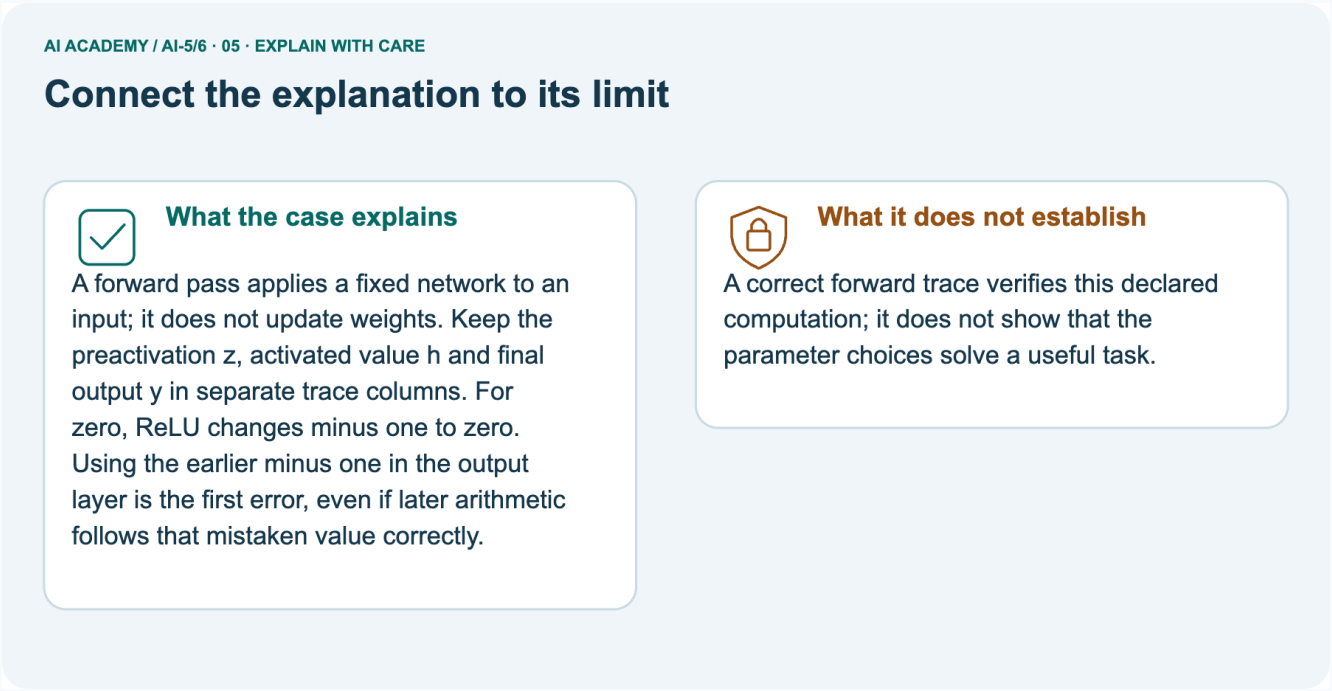
Original case: **Output**s are 4 and 1 respectively. Changed condition: A submitted x = 0 **trace** uses h=-1 and y = 5. Expected result: Its first error is retaining the negative value after ReLU; correct h = 0 gives y = 4.

Reason: ReLU(-1)=0, so the **output** must use 0, not the earlier **preactivation**.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

A forward pass applies a fixed network to an **input**; it does not update weights. Keep the **preactivation** z, activated value h and final **output** y in separate **trace** columns. For zero, ReLU changes minus one to zero.

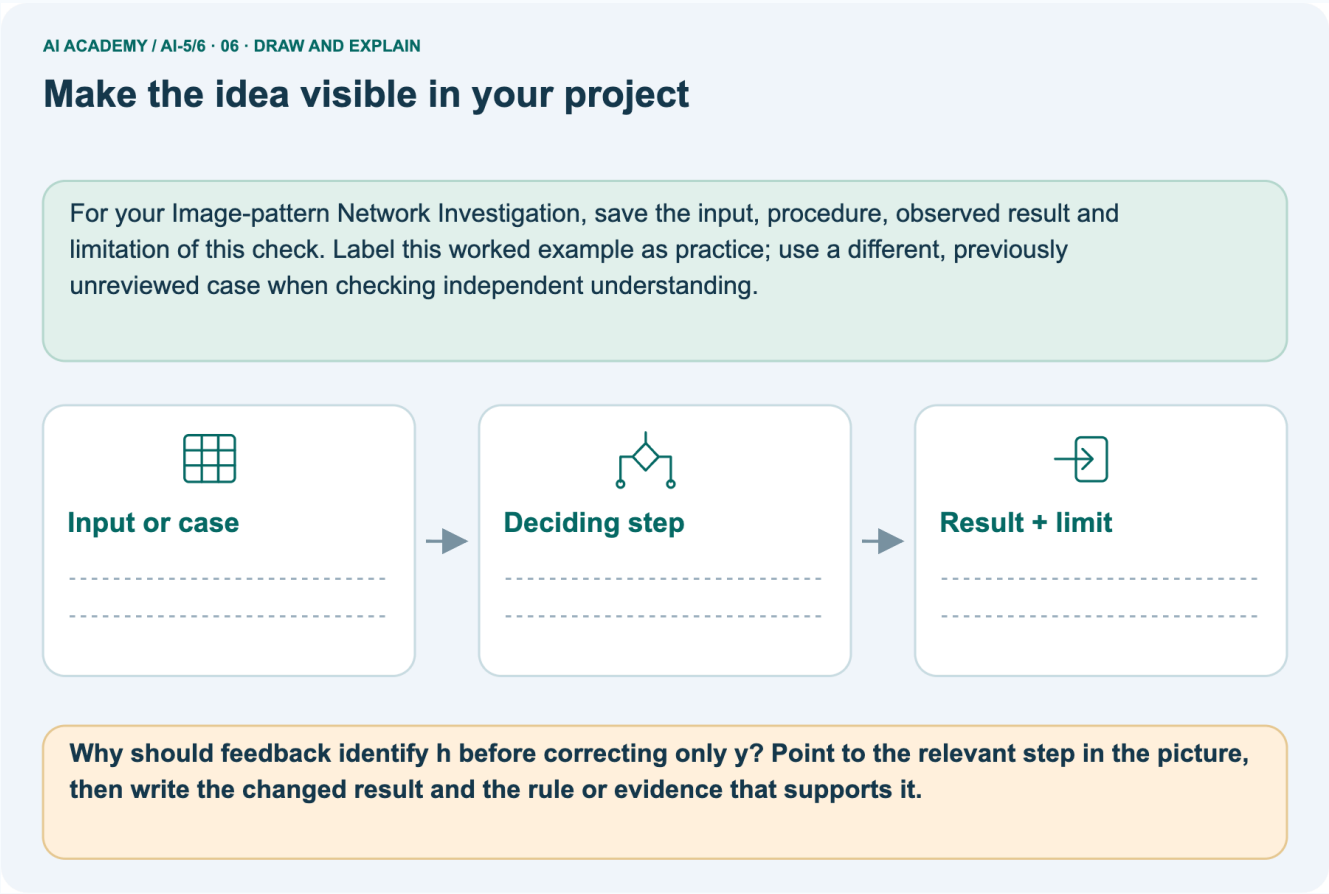
Using the earlier minus one in the **output** layer is the first error, even if later arithmetic follows that mistaken value correctly.

Limit: A correct forward **trace** verifies this declared computation; it does not show that the parameter choices solve a useful task.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Why should feedback identify h before correcting only y? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project **trace** and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. Forward passes by hand: the case

02. Follow the deciding steps

03. Correct a trace from the earliest error

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.